SUPPORTING ADULT GUIDANCE

Putting it together — scale plans

Constructions, Loci and Bearings · Lesson 7. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This capstone lesson draws the whole unit together on scale plans. Learners convert with a scale, choose the correct locus for each rule, and combine several constraints to identify a region — the standard higher-tier 'constructions and loci' problem. It is deliberately synthetic: the mathematics is familiar, and the challenge is selecting and combining the right tools.

Driving Question: *How do constructions, loci and bearings work together to solve a real plan?*

The mathematics, in plain terms

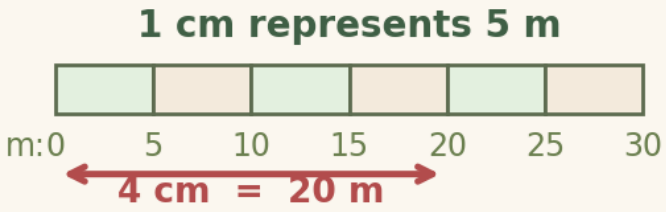
A scale is a fixed ratio between plan and real lengths; plan-to-real multiplies by the scale, real-to-plan divides. Each positional rule is a locus already met in the unit — a circle (fixed distance from a point), a perpendicular bisector (equidistant from two points), an angle bisector (equidistant from two lines), a parallel band (fixed distance from a line). A compound problem is solved by drawing each rule's boundary at the plan scale and shading the region satisfying them all, where the boundaries overlap.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Converting with a scale

A plan is drawn at 1 cm : 5 m. A path measures 4 cm on the plan. How long is it really?

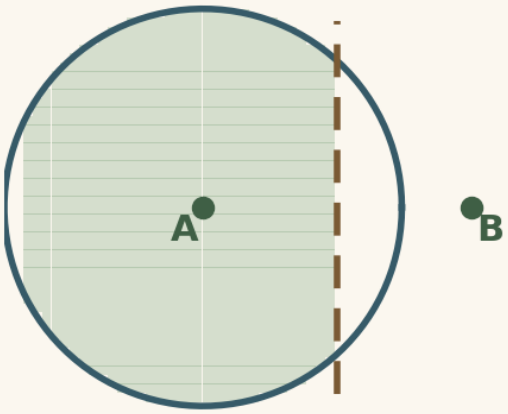


1. The scale is 1 cm represents 5 m.	1 cm : 5 m
2. Plan to real: multiply the plan length by 5.	4 × 5
3. So the real length is 20 m.	= 20 m

20 m — a plan length of 4 cm at 1 cm : 5 m is 4 × 5 = 20 m. (To go back, divide by 5.)

Example 2 — A region from two rules

On a plan, shade where a feature is within a distance of A AND closer to A than to B.



1. 'Within a distance of A' — draw a circle centred A; inside obeys the rule.	circle at A
2. 'Closer to A than B' — construct the perpendicular bisector of AB; A's side obeys it.	perp. bisector
3. Shade where both are true — inside the circle and on A's side.	overlap

The region is the part of the circle on A's side of the perpendicular bisector — where the two loci overlap.

Scale, then the right tool, then the region

Encourage a fixed order: convert distances with the scale first; then, for each rule, name the locus and draw its boundary; only then shade the overlap. The hardest step is *selecting* the right construction, so spend time on 'which locus is this rule?' before any drawing. Insist the scale is stated and construction marks are left showing.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Measure on a scale plan and convert between plan cm and real metres.	~9 min
Movement	Toggle rules on a scale plan and watch the region that obeys them all.	~13 min

Reflection	Match each 'where can it be?' rule to the right locus — the unit's toolkit.	~9 min
Creativity	Solve a real brief: an angle bisector and a circle give an allowed region.	~9 min
Rest	A pause to look back over the whole unit's journey.	~2 min
Nutrition	Name what the learning feeds: synthesis, and the agency of an exact plan.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Multiplying when they should divide (or vice versa) with the scale.
Plan to real multiplies; real to plan divides. Ask: 'Is the real place bigger or smaller than the plan? So do I multiply or divide?'

Picking the wrong locus for a rule.
Ask whether it is 'equidistant from two things' (a bisector) or 'a fixed distance from one thing' (a circle or band), and whether the 'things' are points or lines.

Shading a whole boundary instead of a region.
The answer must obey every rule. Ask the learner to test a point in their shaded area against each rule in turn.

Forgetting to convert distances to the plan scale before drawing.
A 20 m circle at 1 cm : 5 m must be drawn with radius 4 cm, not 20 cm. Ask them to convert first, then set the compass.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the whole unit — bisectors, loci and regions, and bearings. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner draws a 20 cm radius for '20 m' at 1 cm : 5 m, or shades the whole circle for a two-rule problem. The aim is a question that returns the reasoning.

- “What does 1 cm stand for here? So what plan length is 20 m?”

- “Is this rule 'equidistant from two things' or 'a fixed distance from one'? Points or lines?”
- “Pick a point in your shaded region — does it obey BOTH rules? And a point just outside?”
- “Which lesson from this unit does each part of your plan come from?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

How do learners choose a scale?

Often the scale is given. If not, pick one that fits the plan on the page — a round ratio like 1 cm : 5 m keeps the arithmetic simple.

Do they need a protractor and compasses?

Yes — compasses for circles and bisectors, a protractor for any bearing or angle, and a ruler for scale lengths. The Studio mirrors these tools.

What makes a good compound-loci answer?

The scale stated, each boundary constructed correctly (marks showing), and the region satisfying every rule clearly shaded — with the locus named for each rule.

How does this pull the unit together?

Every earlier lesson supplies one tool: equal-distance arcs, rigidity, the two bisectors, loci and regions, and bearings. This lesson is choosing and combining them on a real plan.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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